

General Knowledge

Static GK

STUDY NOTES

Defence Technology



Defence Technology

Introduction

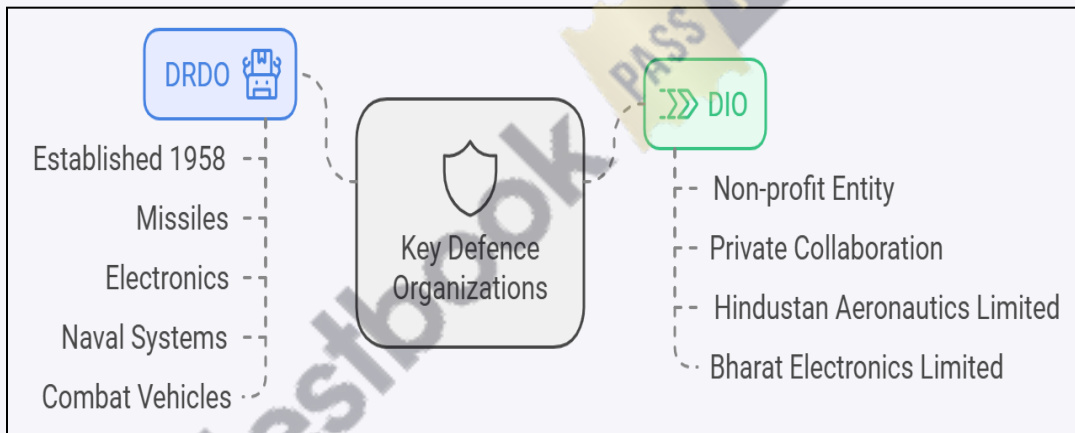
- Defence technology has become a cornerstone of a nation's security apparatus in the modern era, shaping its ability to protect sovereignty and maintain geopolitical balance.
- With its vast and diverse geographical landscape, India faces multifaceted security challenges, including conventional threats from adversarial neighbours and asymmetric threats such as terrorism and cyber warfare.
- In response, India has embarked on an ambitious journey to develop indigenous defence capabilities while integrating cutting-edge global technologies.
- Over the years, India's defence strategy has evolved from dependency on imported systems to becoming a hub of innovation and self-reliance.
- Organisations like the Defence Research and Development Organisation (DRDO) have spearheaded efforts to indigenise critical technologies, reducing import dependency.
- In parallel, strategic initiatives like the Integrated Guided Missile Development Programme (IGMDP) and projects like Mission Shakti have showcased India's growing prowess in missile systems and space defence.
- It also highlights India's approach to maintaining a robust deterrence capability through a triad of nuclear forces, cutting-edge air defence systems, and advancements in unmanned aerial vehicles (UAVs).



Defence Technology: India's Strategic Initiatives, Missile System, & Modernization Efforts Overview:

Organisations

Organisation	Important Information
Defence Research and Development Organisation (DRDO)	Established in 1958, it is the R&D wing of the Ministry of Defence, Government of India.
Defence Innovation Organisation	A 'not-for-profit' company under Section 8 of the Companies Act 2013. Founding members: HAL & BEL (DPSUs).



Defence Research and Development Organisation (DRDO):

- Defence Research and Development Organisation (DRDO) Established in 1958, DRDO is the premier R&D arm of the Ministry of Defence.
- It develops technologies across various domains, including missiles, electronics, naval systems, and combat vehicles.

Defence Innovation Organisation:

- Defence Innovation Organisation (DIO) A non-profit entity under Section 8 of the Companies Act, established to promote innovation and R&D in collaboration with private players like Hindustan Aeronautics Limited (HAL) and Bharat Electronics Limited (BEL).

Committees

Committee	Year	Purpose
N Chandrasekharan	2018	To study the implications of AI in national security.
Shetkar Committee	2015	To suggest measures to enhance combat capability and rebalance defence expenditure.
Defence Artificial Intelligence Project Agency (DAIPA)	2019	Focuses on integrating Artificial Intelligence into Defence Technology.

India's Nuclear Doctrine

Tests Conducted:

- First nuclear test: 1974.
- Second nuclear test: 1998.

Declared Nuclear Weapon State: 1998.

- **Nuclear Doctrine (2003):**
 - Credible Minimum Deterrence.
 - No First Use policy.
 - Nuclear Command Authority: Civilian political leadership authorizes retaliatory strikes.
 - Non-Nuclear Weapon States: Weapons not used against them.
 - Massive Retaliation.

Nuclear Triad

- Land-Based: Agni series (Agni I-V), SLBM (Sagarika K-15), BrahMos.
- Sea-Based: Arihant-class submarines.

- Air-Based: Mig-27.

Missiles:

- Definition: Intelligent unmanned rockets designed to deliver payloads to designated targets.

Type	Features
Solid	Solid fuel, easily storable, achieves high speed quickly.
Liquid	Hydrocarbon-based liquid fuel, complex storage, easily controlled.
Ramjet	An air-breathing engine ignites fuel at subsonic speeds.
Scramjet	Supersonic combustion ramjet, uses hydrogen fuel, combustion at supersonic velocities.
Cryogenic	Liquefied gases (e.g., liquid hydrogen), need insulated containers, and no air intake is required.

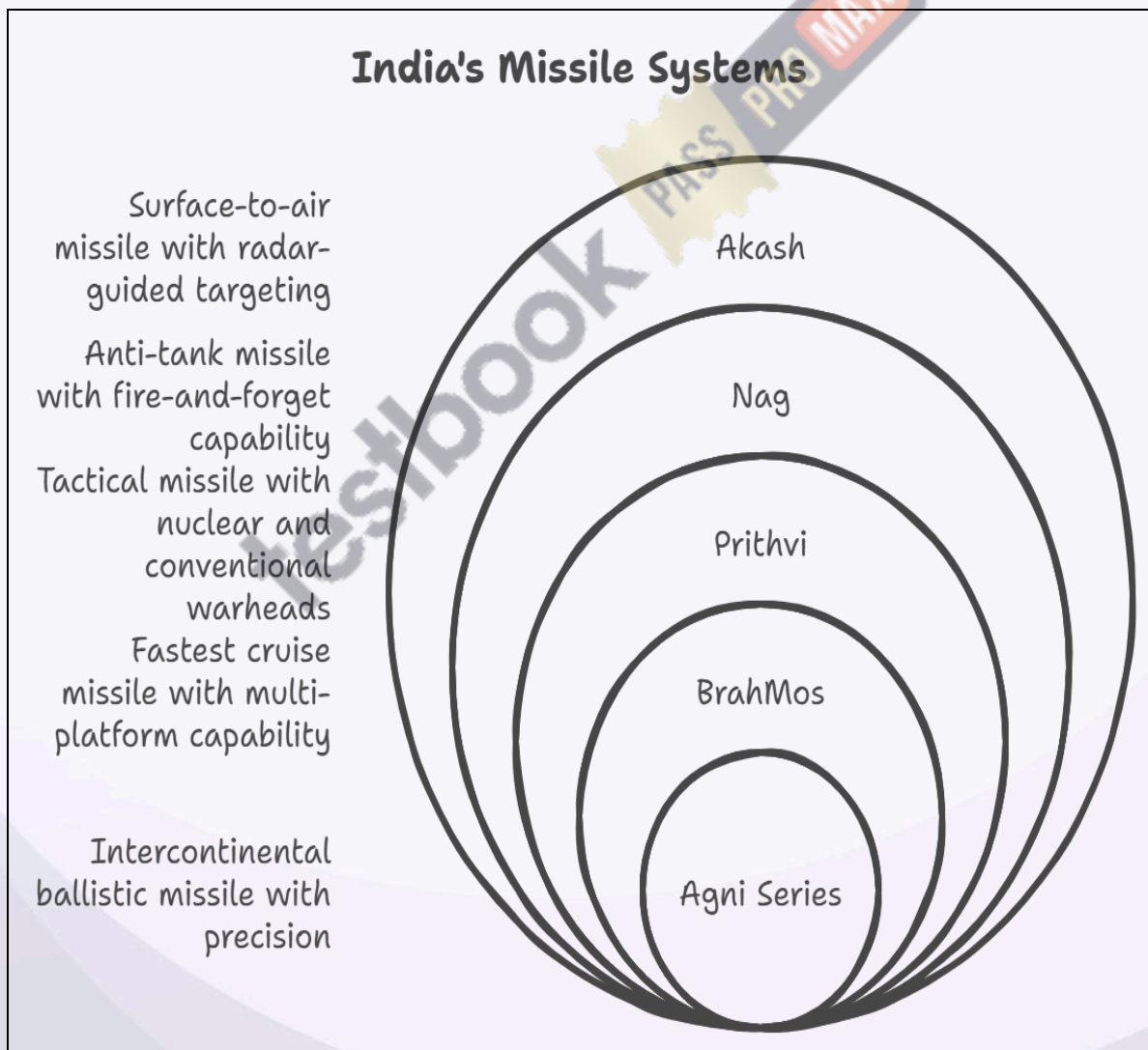
- Missile Types
 - Ballistic Missiles:
 - Propelled briefly after launch, follow a parabolic path.
 - Examples: Prithvi, Agni, Dhanush.
- Range:
 - SRBM (300–1000 km), MRBM (1000–3500 km), IRBM (3500–5500 km), ICBM (>5500 km).
- Cruise Missiles:
 - Self-propelled, low-altitude flight.
 - Examples: BrahMos, Nirbhay.
- Speed:

- Subsonic (< Mach 1), Supersonic (Mach 1-3), Hypersonic (> Mach 5).

Indian Missile Systems

Integrated Guided Missile Development Programme (IGMDP):

- Developed Missiles:
 - Agni: Intercontinental ballistic missile.
 - Prithvi: Tactical surface-to-surface missile.
 - Trishul: Surface-to-air missile, no longer in service.
 - Nag: Anti-tank guided missile.
 - Akash: Medium-range surface-to-air missile.



Missile	Details
Astra	Beyond-visual-range air-to-air missile, 80 km range.
Prahaar	Tactical ballistic missile, 150 km range.
Pralay	Surface-to-surface tactical missile, 350 km range.
Nirbhay	Subsonic cruise missile, 1000 km range, stealth capabilities.
Dhanush	Sea-to-sea ballistic missile, 350 km range.
BrahMos	Supersonic cruise missile, 450–600 km range, multi-platform launch capability.
Pinaka	Multi-barrel rocket launcher, >70 km range.
RudraM-I	Anti-radiation missile, up to 200 km range, targets enemy radar systems.

Anti-Satellite Weapons (ASAT)

Mission Shakti:

- Tested in March 2019.
- Destroyed a satellite in low-earth orbit.
- Demonstrates India's space-superpower status.
- Air Defence Systems

Ballistic Missile Defence (BMD):

- A two-tiered system with Prithvi Air Defence (PAD) for high-altitude and Advanced Air Defence (AAD) for low-altitude interceptions.
- Systems:

- S-400 Triumf (Russia): Mobile, surface-to-air missile system.
- Iron Dome (Israel): Short-range missile interceptor.

Navy

• Submarine Types:

- Nuclear-powered: Long endurance, second-strike capability.
- Diesel-electric: Limited endurance, easy to detect.
- Attack Submarines: Designed for combat tasks.
- Ballistic Missile Submarines: Launch nuclear warheads.

Projects:

Project	Features
Project 75	Six conventional submarines with AIP (e.g., INS Kalvari).
Project 28	Four anti-submarine warships were indigenously built (e.g., INS Kamorta).
Project 17A	Seven stealth frigates.

Modernization Initiatives

- Strategic Partnership Model: Indian private companies tie up with global OEMs.
- iDEX Initiative: Promotes innovation in defence by engaging startups and academia.

Defence Indigenisation:

- Increased FDI cap to 74%.
- SRIJAN portal for private-sector participation.
- Defence industrial corridors in Tamil Nadu and UP.

Acquisition Policy (DAP 2020)

- Enhanced Indigenous content requirements.
- New leasing category for acquisitions.

Drones

- Categories: Nano, Micro, Small, Medium, Large.
- Drone Regulations: The NPNT clause mandates regulatory permission.

Chemical and Biological Weapons

- Chemical Weapons:
 - India ratified the Chemical Weapons Convention (CWC) in 1996.
 - Common agents: Chlorine gas, Sarin, VX.
- Biological Weapons:
 - Examples: Anthrax, Plague.
 - Biological Weapons Convention (BWC) bans development and stockpiling.